

8

Quality of Service and High Availability

“Quality is never accidental; it is always the result of intelligent effort.”

John Ruskin

- Understanding the concept of QoS and its objectives, as well as its usefulness for current IT infrastructures.
- Understanding the basic principles of QoS in networks (needs, structure, configuration options).
- Learning how to use data flow classification, improved enqueueing, and advanced load management techniques.
- Differentiating between QoS protocols or standards in the link and network layers.
- Understanding high availability mechanisms, in this case redundancy and backup.
- Describing the objectives and main protocols of level 2 and 3 redundancies.
- Understanding the general functioning of load balancing and data replication.

8.1. Introduction

These days, IT infrastructures offer more and more diverse services and serve as a framework for a wide range of applications. This varied offer must be accompanied by a service quality of assurance in order to ensure end-user satisfaction.

In reality, and in a context of convergence, services with different characteristics and different requirements must be provided by a single infrastructure that must ensure their proper functioning consistent with the levels of service required, or SLA (Service Level Agreement). Therefore, any plan to set up a network infrastructure must include and address the service quality aspect.

In the same vein, an effort to achieve high availability of the various components of the network infrastructure, mainly through redundancy, is certainly in order. This high availability helps to avoid crashes and to balance loads in order to optimize the functioning of the network infrastructure.

8.2. Quality of service

8.2.1. *Motivation*

Quality of Service (QoS) played a secondary role in the original design of the TCP/IP protocols, designed to transport data from asynchronous applications, that is, services

such as file transfers, e-mail and terminal access.

However, some applications, such as VoIP and Video IP, require an adequate network QoS: controlled latencies, low jitter and low packet loss, as well as guaranteed bandwidth. This is the only way quality is guaranteed for the end-user. QoS relies on defined control mechanisms that are linked to traffic management in order to influence transport network metrics in relation to flows of information transmitted.

QoS is not a supplementary functionality like an extension for a network infrastructure, then, but the result of a large number of coordinated measures that must be anchored in the basic design of a network infrastructure. It is important to note here the interaction of the various layers of the OSI model and their requirements; indeed, it is useful to determine whether it is possible to have mechanisms in a layer that enable tolerance against malfunctions in the underlying layers.

The methods required will be discussed in this chapter. The functional elements of a network that are necessary for good QoS, and how they can be implemented with current technologies, will also be presented.

Prioritization or configuration of data traffic related to QoS, reservation of data output, limitation of data output and packet optimization are carried out by multiple mechanisms such as virtual local area networks (VLANs), DiffServ and IntServ architectures, traffic shaping, etc.

8.2.2. Definition(s)

There are various definitions of the term “Quality of Service”. These include the following:

– Generic definition

“Quality in general has to do with consistency between a subject's expectations regarding the properties of an object.”¹

Quality of service (QoS) refers to the well-defined, manageable behavior of a system according to measurable parameters.

– According to Cisco

QoS refers to the capacity of a network to provide improved service for network traffic based on various underlying technologies, including Frame Relay, ATM, Ethernet, and 802.1 and SONET networks, etc.

– According to IUT-T

All of the characteristics of a telecommunications service having to do with its capacity to satisfy the stated and implicit needs of the user of the service.

– According to Wikipedia

QoS is the overall performance of a telephone or IT network, particularly as perceived by network users. To measure QoS quantitatively, various aspects of network service are often taken into account, such as error rate, bandwidth, output, transmission time, availability, jitter, etc.

QoS can thus be considered part of the characteristics of a service offered by a network. The network itself may offer various service models. It is within this context that the SLA falls. The SLA is contractually defined between the service provider and the client for a specific application based on the service models offered.

Thus, QoS is an important means of assessing and judging the usefulness and above all the performance of teleinformatic networks and their services.

8.2.3. Objectives of QoS

QoS is principally aimed at remedying problems of network congestion or saturation, which cause latencies and data loss. It also enables the optimal support of multimedia applications, which are based on the principle of real time. Transactional applications require reasonable response time and can therefore benefit from QoS. In fact, QoS abandons the principle of flow neutrality, according to which all frames (layer 2) and packets (layer 3) are processed according to the “first come, first served” principle, or “FIFO” in switches and routers.

QoS functionality also makes it possible to achieve the following objectives:

- ensuring a certain level of quality for one or more services (the VoIP service, for example);
- isolating/protecting a specific flow from other flows;
- controlling the distribution of network resources.

The two main approaches used to achieve these objectives are:

- oversizing the network by providing a greater bandwidth in order to deal with peak traffic times. However, this solution is not economically viable, and involves poor management of resources. This is in addition to the fact that a network's capacity cannot be infinite;
- developing methods to distinguish between different traffic flows in order to classify them according to their bandwidth requirements and prioritize them according to their criticality. In sum, the goal is to manage network resources intelligently.

8.2.4. Metrics of QoS

In order to evaluate or measure QoS as well as the impact of the various underlying mechanisms, we use quantifiable parameters that characterize any data flow, specifically bandwidth, latency, jitter and packet loss, as recapitulated in [Table 8.1](#).

Table 8.1. *QoS parameters*

Parameter	Definition	Example
Bandwidth	End-to-end output available on the network, which is the total number of IP packets successfully transferred in a given period of time.	2 Mbps
Latency	The time necessary for a packet to traverse the network from end to end, from the transmitter to the receiver, on a given support.	100 ms
Jitter	Variation in end-to-end transit time between two consecutive packets.	If the transit time varies between 80 and 120 ms, jitter is 40 ms.
Packet loss	The ratio of the number of packets lost to the number of packets transmitted during transmission.	1%

Each type of traffic requires a specific cluster of parameters.

Table 8.2. *QoS parameters for voice and video*

Traffic type	Required parameter cluster
VoIP	Output = 64 kbps Latency < = 150 ms (to be verified) Jitter < = 30 ms Loss rate = 1%
Video, video conferencing, streaming	Output = From 384 kbps to more than 20Mbps Latency = 200 to 400 ms Jitter = 30-50 ms Loss rate = 0.1-1%

In addition to the parameters cited above, which present measurable and quantifiable parameters, an assessment of QoS as perceived by the user can be useful. Thus, the QoS of a voice communication can be assessed through an MOS (Mean Opinion Score). MOS value ranges from 1 (unacceptable) to 5 (excellent) and is obtained from subjective test results.

8.2.5. General principles of QoS

QoS parameters are dependent on connection capacity and dynamic queuing management within switch nodes (or routers). If the network shows an overload, taking into consideration the fact that each type of traffic has different requirements in terms of QoS parameters, a queue for each class of traffic is required. The routers

involved in a network implementing QoS must provide packet classification functions in sufficient queues, management of packets placed in the same queue and scheduling to control sharing of the finite capacity of the link among multiple traffics of different types.

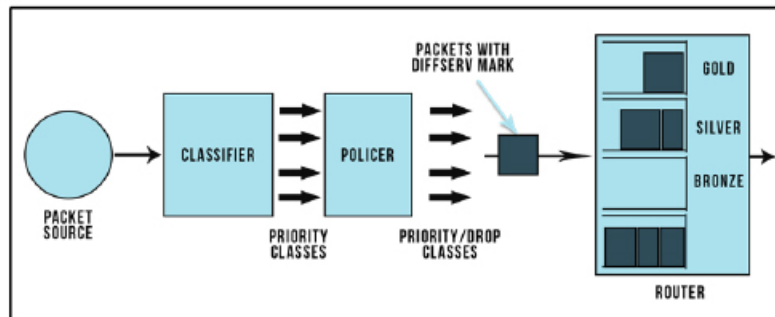


Figure 8.1. General principles of QoS establishment. For a color version of the figure, see www.iste.co.uk/helali/systems.zip

8.2.5.1. Classification

Classification is the first stage of QoS implementation. The principle consists of analyzing a packet to determine the corresponding flow, thus determining the appropriate level of service. The classification function is carried out by a flow or packet classifier based on specific classification rules. These rules may refer to headers or tagging, ports, source and destination addresses, etc. This stage is of vital importance, as it enables packet differentiation.

The packet tagging mechanism enables routers to distinguish between the different classes of data flows so that the equipment crossed is not forced to reclassify the packets received each time.

8.2.5.2. Queuing management

Queues serve mainly to manage congestion in a network infrastructure. A corresponding queue is created for each tag or class.

Managing these queues involves mechanisms to manage buffers, a resource limited at the switch or router level. The objective is to control the number of packets in a queue by deciding which packets to reject in the event of congestion. This means it is a packet loss control function. There are multiple techniques corresponding to this, including last-packet rejection and random early detection (RED), in which packets are randomly rejected before the queue reaches a predefined threshold. An improved variant of RED, weighted random early detection (WRED), has been developed by Cisco, which defines multiple thresholds according to traffic class.

8.2.5.3. Scheduling

Scheduling involves determining packet processing order at the router output level. This decision, based initially on the principle of equity, can favor certain flows to the detriment of others (real-time applications, certain protocols, specific machine source, etc.). This functionality has a significant effect on delay and output parameters for each flow. In this way, the scheduler controls the access of service classes to a limited

network resource, namely bandwidth. Several scheduling techniques or algorithms can be applied, including but not limited to:

- strict priority: a priority n queue is fully processed before moving to priority $n-1$ queue;
- round robin priority: the router processes a packet from each queue beginning with the highest-priority queue;
- weighted priority or weight fair queuing (WFQ), in which each output queue receives a certain weighting. The available bandwidth is then shared among the different flows according to the weighting assigned.

8.2.5.4. Traffic policing/shaping

These are mechanisms to shape and regulate traffic, thus making it possible to control the output or bandwidth of each traffic segment in order to reduce traffic burstiness.

Traffic shaping, an operation carried out at input or within a network, ensures that traffic remains within a limit value. It helps to avoid peaks and provides a quasi-constant flow. Excessive traffic is placed in a buffer and released when possible.

Traffic policing at network input processes traffic that exceeds limits in a different way; it generally eliminates or (re)tags these packets without introducing latency into the network.

Implementing QoS in an IT infrastructure requires a preparatory phase that consists of:

- identifying types of traffic in the infrastructure and their requirements;
- determining the importance of each type of traffic for optimal company activity;
- classifying traffic;
- defining the policy to be applied to each class (minimal guaranteed bandwidth, maximum bandwidth, fixed priority level);
- defining the QoS mechanisms to be adopted to manage congestion (policing, shaping).

8.2.6. QoS mechanisms

QoS is a key factor for managing multimedia applications and for the success of the IP protocol as a basic unifying protocol for next-generation convergent networks. To that end, various solutions or mechanisms have been proposed to differentiate among network flows and process them adequately.

These mechanisms are based on two main approaches:

- either adapting application flows to network capacity; or
- adapting the network to application flows by managing flows via one or more resources by grouping them by class.

8.2.6.1. *In the transport layer*

At this level, RTP and RTCP protocols are implemented to manage real-time applications such as VoIP and streaming. They represent a concretization of the first approach. In other words, they make it possible to adapt the flow of an application to the bandwidth available from the network:

- data transmission in RTP (Real-Time Protocol) packets. RTP makes it possible to number packets, thus enabling the detection of lost packets, media synchronization and identification of a message's content for possible recovery in case of loss. RTP transports user data;
- quality control of the session (output, packet loss, jitter, latency) is managed by RTCP (Real-Time Control Protocol) packets. This protocol is used by receivers to send a report containing the main service quality parameters to transmitters. This information enables the source to adapt to the situation. In this way, the RTCP transports supervision packets, enabling network nodes to better control real-time applications.

8.2.6.2. *In the network layer*

This involves adapting the network to the characteristics of application flows by managing individual flows or groups of classified flows. The IETF has released both a best effort model for this and two network layer QoS management models.

IntServ

IntServ processes each flow separately from the others. To do this, it uses a specific signaling protocol called RSVP (Resource ReSerVation Protocol).

RSVP (Resource Reservation Protocol) is used to reserve resources. It is used by hosts requiring well-defined QoS for specific flows as well as the routers located along the path of these flows. Memory resources on the crossed routers are reserved by the source via signaling messages. QoS modification is possible following a request by the receiver. QoS is not integrated into each packet, but is negotiated with network equipment; this is what we refer to as end-to-end QoS. RSVP is a signaling or control protocol and does not transport user data.

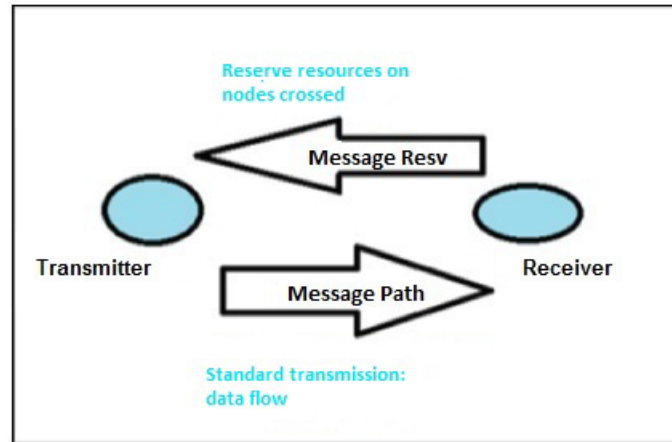


Figure 8.2. Basic principles of RSVP. For a color version of the figure, see www.iste.co.uk/helali/systems.zip

This IntServmodel ensures fine granularity and isolates each flow from the others. However, it is relatively complex and costly. It also comes with problems of scalability (resizing to a large scale). It can be deployed on intranets (LANs) but would be difficult to use on the scale of a service provider, for example.

DiffServ

DiffServ is based on differentiating between packets according to their class. QoS is managed by class and represents an aggregation of multiple flows. Thus, multiple classes with different qualities of service are defined by means of packet tagging from the source. Once a packet has been classified and tagged by the switch or router, a type of processing called per-hop behavior (PHB) is applied to it. PHB identifies the processing of a packet at each hop all along the path from the source to the destination. Three types of PHB are currently used:

- PHB by default, which corresponds to the best effort model;
- EF (Expedited Forwarding), which ensures that each node configured with DiffServ guarantees minimum values for latency, jitter and packet loss. This guarantee is necessary for real-time applications such as voice and video;
- AF (Assured Forwarding) provides lower quality levels than EF and is suitable for non-real-time applications.

Tagging consists of modifying the headers of IPv4 packets entering the TOS (Type of Service) field, replaced later by the DSCP (Differentiated Service Code Point) field as shown in [Figure 8.3](#), as well as the headers of IPv6 packets and, more specifically, the traffic class field (see [Figure 8.4](#)).

Version (4 bits)	Header length (4 bits)	Service type TOS or DSCP (8 bits)	Total length (16 bits)
Identification (16 bits)		Flag (3 bits)	Offset Fragment (13 bits)
Lifespan (8 bits)	Protocol (8 bits)	Checksum (16 bits)	
IP source address (32 bits)			
IP destination address (32 bits)			
Options (possible)			

Figure 8.3. IPv4 header and TOS/DSCP fields. For a color version of the figure, see www.iste.co.uk/helali/systems.zip

Version (4 bits)	Traffic Class (8 bits)	Flow identification label (20 bits)	
Length of useful data (16 bits)		Following header (8 bits)	Hop count (8 bits)
IP source address (128 bits)			
IP destination address (128 bits)			

Figure 8.4. IPv6 header and traffic class field. For a color version of the figure, see www.iste.co.uk/helali/systems.zip

The TOS field (IP precedence) displays as follows.

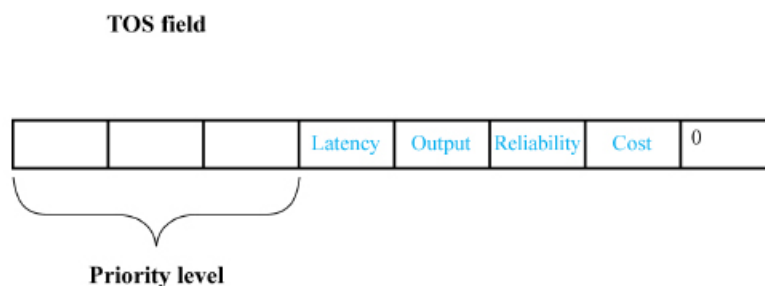


Figure 8.5. Structure of TOS (Type Of Service) field

The DSCP field is coded on 6 bits and makes it possible to increase the number of priority levels compared to the TOS field, which is coded on only 3 bits.

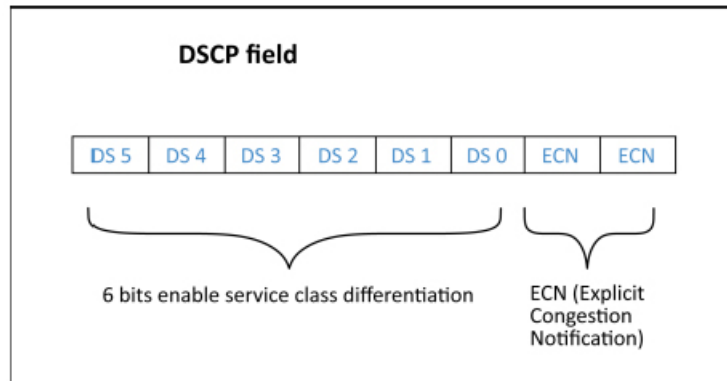


Figure 8.6. *DSCP (Differentiated Service Code Point) field structure*

The DiffServ model, by basing itself on the DSCP field, offers higher granularity, refining it by adding the 3 bits DS2, DS1 and DS0. The first three bits, DS5, DS4 and DS3, define the class called IP precedence just as they do in the TOS field. The bits DS2 and DS1 designate loss probability, while DS0 is always set at 0.

IP precedence uses the three most significant bits of DSCP, thus ensuring compatibility with the previous technique or TOS. However, it allows only eight levels of classification. Values 6 and 7 are reserved for network control and routing protocols. IP voice uses value 5, video and video conferencing value 4. The default value, or 0, is recommended for best effort traffic.

Network border routers are responsible for classifying and tagging incoming packets according to several possible criteria (header, VLAN, IP addresses, ports, ACL class-map, etc.), while routers within the network process these packets differently depending on their tag or class.

8.2.6.3. In the data link layer

The IEEE 802.1p method represents a QoS process defined in the IEEE 802.1Q standard. This technique uses a 3-bit subfield in the header of the Ethernet frame to assign different priority levels. 802.1p traffic priorities are thus included in VLAN tags. The objective is to differentiate between the types of traffic. Network devices compliant with the 802.1p standard can classify data traffic into priority classes. In the event of network congestion, higher priority frames are transmitted before lower priority ones.

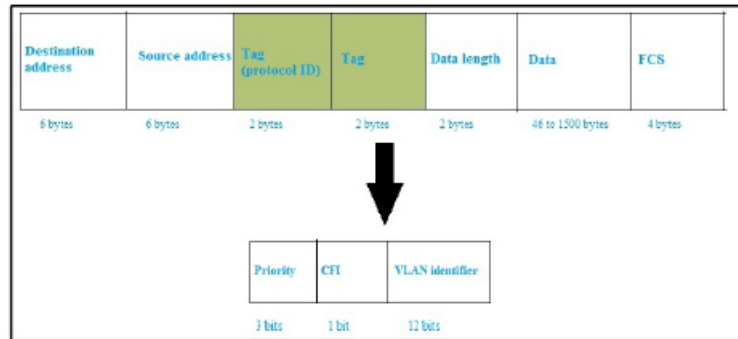


Figure 8.7. Structure of IEEE 802.1q frame. For a color version of the figure, see www.iste.co.uk/helali/systems.zip

With the three bits, seven classes of service are offered. These are shown in [Table 8.3](#).

Table 8.3. Different priority categories

Decimal value of three bits	Priority categories
0	Best effort
1	Background
2	Reserved
3	Excellent effort
4	Streaming, multimedia
5	Interactive video
6	Voice
7	Network control reserved

QoS configuration for routers and switches is possible via several mechanisms provided by Cisco, for example:

- through the CLI command line: definition of QoS strategies at interfaces. This process is difficult to control and non-scalable;
- AutoQoS: this is a simple mechanism that ensures consistent policy application to all equipment;
- QoS Policy Manager (QPM): this mechanism implements QoS on the scale of an Internet access provider, for example;
- MQC Modular QoS CLI (Command Line Interface): this is based on modularity, and indeed QoS is constructed in blocks or modules.

8.2.6.4. Applications

Application 1:QoS configuration for routers

By default, routers with no QoS mechanisms process all packets in the same way, according to the first-come-first-served (FIFO) principle, but this behavior can corrupt the operation of certain types of critical traffic.

Here, we will use the MQC (*Modular QoS Command Line Interface*) approach, which has a number of advantages:

- standardization of the implementation of QoS on all Cisco platforms while masking the underlying hardware architecture. In this way, clients can upgrade to the new generation of Cisco hardware and copy existing MQC configurations;
- distinguishing between classification modifications and strategy modifications, and deactivating a QoS strategy with a single command, thanks to modular configuration. One QoS strategy can also be applied to multiple interfaces.

This approach is concretized via a three-stage process:

1) Definition of a flow class or class-map. This stage defines a set of network traffic, or a class-map in MQC terminology, using several classification criteria: protocols, access control lists (ACL). IP addresses, IP precedence, DSCP, IEEE 802.1p, etc. The class name must be specified as well as a series of match commands used to set classification criteria.

The command `router(config)#class-map match-all classname` stipulates that the packet must satisfy ALL criteria to belong to the class.

The command `router(config)#class-map match-any classname` indicates that the packet must satisfy at least one of the criteria set in order for it to belong to a class.

Once the service class has been created, we move on to specifying the criteria themselves. [Table 8.4](#) gives a few examples of this.

Table 8.4. *Examples of classification criteria*

Criterion	Explanation
<code>router(config-cmap)#match input-interface interface name</code>	The packet must come from the input interface set in order to belong to the corresponding class
<code>router(config-cmap)#match access-group ACL number</code>	The corresponding ACL determines whether the packet belongs to the class
<code>router(config-cmap)#match protocol protocol name</code>	The protocol determines whether the packet belongs to the class

2) Definition of a QoS policy or policy-map in which each flow class is assigned a priority level. The definition of a policy can be used for tagging as well as bandwidth sharing. This stage represents the effective implementation of the QoS set according to the company's needs. In concrete terms, the class-map will be used to apply QoS functions such as tagging, queuing, suppression, regulation and shaping.

3) Application of the policy developed at the corresponding interface, either incoming or outgoing according to the need.

The infrastructure schema adopted for this example is given in [Figure 8.8](#).

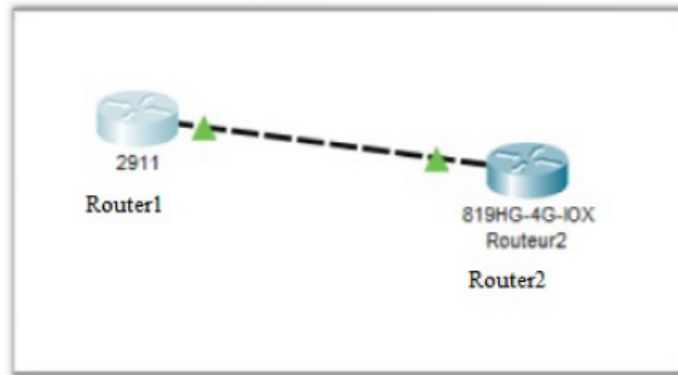


Figure 8.8. *Topology of our example*

Stage 1: creation of class-map or flow classes

The first class-map to be defined concerns VoIP traffic:

```
Router1(config)#class-map VOICE
Router1(config-cmap)#match protocol rtp audio
Router1(config) #exit
```

The second class-map defined is dedicated to web traffic:

```
Router1(config)#class-map match-any WEB
Router1(config-cmap)#match protocol http
Router1(config-cmap)#match protocol https
Router1(config) #exit
```

The third class-map to be defined pertains to FTP traffic:

```
Router1(config)#class-map match-any TRANSFER
Router1(config-cmap)#match protocol ftp
Router1(config-cmap)#match protocol tftp
Router1(config) #exit
```

The final result is recapitulated below:

```
Router1#show class-map
Class Map match-any class-default (id 0)
Match any
Class Map match-any TRANSFER (id 1)
Match protocol ftp
Match protocol tftp
Class Map match-all VOICE (id 2)
Match protocol rtp
```

Class Map match-any WEB (id 3)

Match protocol http

Match protocol https

Stage 2: creation of policy-map

A policy-map called QOSPOLICY is created using the following commands:

```
Router1(config)#policy-map QOSPOLICY
```

```
Router1(config-pmap)#class TRANSFER
```

```
Router1(config-pmap-c)#bandwidth percent 10
```

```
Router1(config-pmap-c)# random-detect dscp-based
```

A policy-map called QOSPOLICY has now been created. Next, we begin with the TRANSFER class-map. With the bandwidth command, we assign at least 10% of the available bandwidth to it. The command random- detectdscp-based, an active WRED (Weighted Random Early Detection) technique specific to queue management in a router, sets the parameters for the decision to suppress packets according to their DSCP values.

Next, we enter policy-map-class configuration mode to address the VOICE traffic class and assign its priority:

```
Router1(config-pmap)#class VOICE
```

```
Router1(config-pmap-c)#priority 256
```

This traffic class will now have a maximum bandwidth of 256 kbps but no more, and its packets will be sent before other packets.

Finally, we configure the WEB traffic class:

```
Router1(config-pmap)#class WEB
```

```
Router1(config-pmap-c)#bandwidth percent 60
```

```
Router1(config-pmap-c)#random-detect dscp-based
```

In this way, web traffic benefits from at least 60% of bandwidth, and packet suppression is based on the value of the DSCP field.

Stage 3: application of policy-map

We apply the QOSPOLICY policy at the gio/o interface in the output direction:

```
Router1(config)#interface gio/o
```

```
Router1(config-if)#service-policy output QOSPOLICY
```

Application 2: QoS configuration for switches (general principles)

The implementation of QoS in this application is based on the DiffServ model, which specifies that each packet be classified upon entering the network.

Configuration commands can vary from one manufacturer to another. This configuration example is based on the Cisco Catalyst 3750 switch.

Classification, and subsequently prioritization, can be based on level 2, the IEEE.1q frame header and, more precisely, the COS field (COS values vary from 0 to 7). They can also refer to level 3, the IP packet header, and more precisely the TOS field, which can involve IP precedence or DSCP. QoS manages the use of one value or the other, as DSCP values are compatible with the previous versions of IP precedence. IP precedence values range from 0 to 7, while DSCP field values range from 0 to 63.

The class information contained in the frame or packet can be assigned by end hosts, switches or routers along the pathway, according to a configured strategy, a detailed examination of the packet, or both. A detailed examination of the packet to be carried out close to the network edge so that the switches and routers at the heart of the network are not overloaded by this task.

By default, QoS is deactivated in Cisco Catalyst 3750 switches. In other terms, incoming frames retain their original QoS tags. The switch does not modify COS or DSCP values:

```
Switch# show mlsqos
```

QoS is disabled

QoS ip packet dscp rewrite is enabled //Traffic will be routed with
the same COS and DSCP values as at input despite this assignment

The command used to activate QoS functionality is:

```
Switch(config-if)#mlsqos trust [ cos / device / dscp / ip-precedence ]
```

cos cos keyword

device trusted device class

dscpdscp keyword

ip-precedence ip-precedence keyword

The COS value (3 802.1p priority bits) and the DSCP value of the frame entering the switch have a default value of 0. The DSCP value is calculated according to the COS–DSCP correspondence table.

Classification and tagging of incoming frames according to COS/DSCP is done in three ways: port-based configuration, MQC configuration or VLANbased configuration. We will now discuss in detail QoS configuration at ports for a Cisco Catalyst 3750 switch.

Configuration according to ports

Configuration is based mainly on the principle of trust in the original tags of incoming frames or packets.

In the negative case, the packet will be classified using a QoS tag based on the access list (ACL) or tagging. Otherwise, there are several possibilities:

First case: in the case of an access port or a level 3 port, the command to use will be:

```
Switch(config-if)#mlsqos trust dscp
```

NOTE.- the mlsqos trust cos command cannot be applied in this case due to the lack of IEEE 802.1q or ISL tag.

Second case: involves a trunk or tagged port.

In this case, we can use one of the two following commands:

```
Switch(config-if)#mlsqos trust cos
```

or

```
Switch(config-if)#mlsqos trust dscp
```

MAP DSCP/COS table is used to calculate the value of COS if the port is configured based on the DSCP field. The MAP DSCP/COS table is also used to calculate the DSCP value if the port is configured to trust the COS.

Third case: involves a trunk or tagged port configured with the command `mlsqos trust cos`.

Frames coming from the native VLAN will have COS and DSCP values defined at 0. However, it is possible to configure the switch port to modify the COS value for non-tagged frames using the command:

```
Switch(config-if)#mlsqos cos [1..7]
```

With this command, the port trusts the COS values of tagged frames and defines the COS value of non-tagged frames at 0.

If the port is configured with the command:

```
Switch(config-if)#mlsqos cos 3 override
```

then the port defines the COS values of all frames (tagged or not) at 3. Previously configured trust values are thus disregarded.

Fourth case: a port in access mode to which a PC and an IP telephone are connected.

The following command is used to tell the Cisco IP telephone to tag data traffic coming from a PC connected to the same port with a predetermined class of service (COS) value. The objective of this is to prevent the PC from tagging its frames with the VLAN voice tag and thus not assigning the same priority to voice and data traffic:

```
Switch(config-if)#switchport priority extend cos [0..7]
```

It is now possible for a user to tag the PC data with a dot1q frame with a higher COS value. To avoid this situation and keep giving higher priority to voice traffic, the following command can be used:

```
Switch(config-if)>mlsqos trust device cisco-phone
```

After classification comes tagging. Mapping tables between COS and DSCP and vice versa are created with default values and then activated. For example, the following command gives the correspondence between COS values and (default) DSCP values:

```
Switch# shmlsqos maps cos-dscp
```

Cos-dscp map:

```
cos: 0 1 2 3 4 5 6 7
```

```
dscp: 0 8 16 24 32 40 48 56
```

The COS-DSCP correspondence table is shown in [Table 8.5](#).

Table 8.5. *COS-DSCP correspondence table*

COS	DSCP (decimal)	DSCP
0	0	default
1	8	CS1
2	16	CS2
3	24	CS3
4	32	CS4
5	40	CS5
6	48	CS6
7	56	CS7

To manage VoIP traffic, the required DSCP value is EF (Express Forwarding), as shown in [Figure 8.9](#) in the first line.

In our case, the port is configured to trust the COS value (mlsqos trust cos). Here, the switch calculates the DSCP value CS5 (40) to obtain COS value 5 from the COS-DSCP table. Voice traffic is thus identified by the DSCP value CS5. This is not the desired value. To obtain the DSCP value EF, we use the following command:

Switch(config)# mlsqos map cos-dscp 0 8 16 24 32 40 48 56

Application Class	Media Application Examples	PHB
VoIP Telephony	Cisco IP Phone	EF
Broadcast Video	Cisco IPVS, Enterprise TV	CS5
Real-Time Interactive	Cisco TelePresence	CS4
Multimedia Conferencing	Cisco CUPC, WebEx	AF4
Multimedia Streaming	Cisco DMS, IP/TV	AF3
Network Control	EIGRP, OSPF, HSRP, IKE	CS6
Call-Signaling	SCCP, SIP, H.323	CS3
Ops/Admin/Mgmt (OAM)	SNMP, SSH, Syslog	CS2
Transactional Data	ERP Apps, CRM Apps	AF2
Bulk Data	E-mail, FTP, Backup	AF1
Best Effort	Default Class	DF
Scavenger	YouTube, Gaming, P2P	CS1

Figure 8.9. Examples of the application and corresponding DSCP values. For a color version of the figure, see www.iste.co.uk/helali/systems.zip

The result is therefore

Table 8.6. *Modified COS-DSCP correspondence table*

COS	DSCP (decimal)	DSCP
0	0	default
1	8	CS1
2	16	CS2
3	24	CS3
4	32	CS4
5	46	EF
6	48	CS6
7	56	CS7

[Table 8.7](#) recapitulates the correspondences between COS and DSCP values.

Table 8.7. *Complete COS-DSCP correspondence table*

COS	DSCP (decimal)	DSCP
0	0-7	default
1	8-15	CS1 AF11 AF12 AF13
2	16-23	CS2 AF21 AF22 AF23
3	24-31	CS3 AF31 AF32 AF33
4	32-39	CS4 AF41 AF42 AF43
5	40-47	CS5 E-F
6	48-55	CS6
7	56-63	CS7

NOTE.— All network switches must have identical correspondence tables to avoid QoS configuration problems.

Application 3: QoS configuration at a level 3 switch with VoIP management

This example of configuration gives the procedure for setting up a switch for better VoIP management based on the detection of VLAN voice and the prioritization of its flows.

There are two possible architectures. In the first, the IP telephone and the PC are connected to two separate ports. In the second architecture, they are connected to the same port.

We begin with the creation of two VLANs, voice and data, as shown below:

```
Switch(config)#vlan 10
```

```
Switch(config-vlan)#name voice
```

```
Switch(config)#vlan 20
```

```
Switch(config-vlan)#name data
```

First architecture

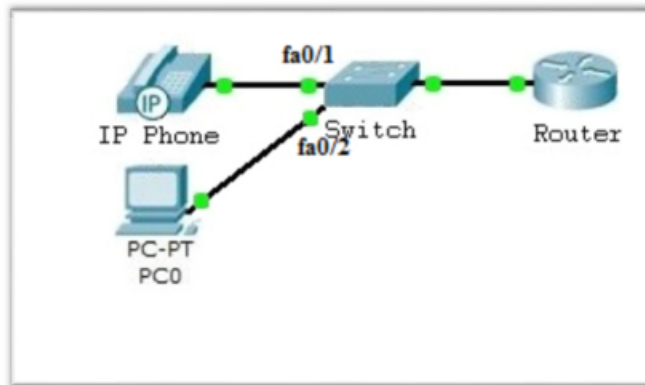


Figure 8.10. *Implementation of a VoIP solution (first architecture)*

In this case, a different configuration must be applied for the port that transports VoIP (interface fa0/1) and the port that transports normal data, fa0/2.

Second architecture

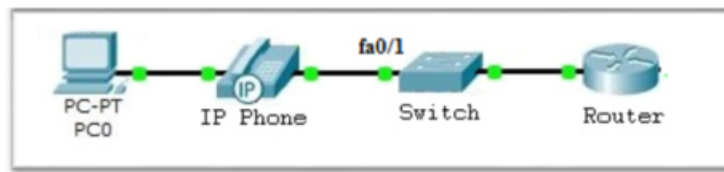


Figure 8.11. *Implementation of a VoIP solution (second architecture)*

This architecture is more advisable as it saves cabling and the application of only one configuration for a single port (fa0/1).

The CDP protocol (Cisco Discovery Protocol) is used to enable telephones to situate themselves in the right VLAN automatically.

The topology shown in [Figure 8.12](#) encompasses both architectures.

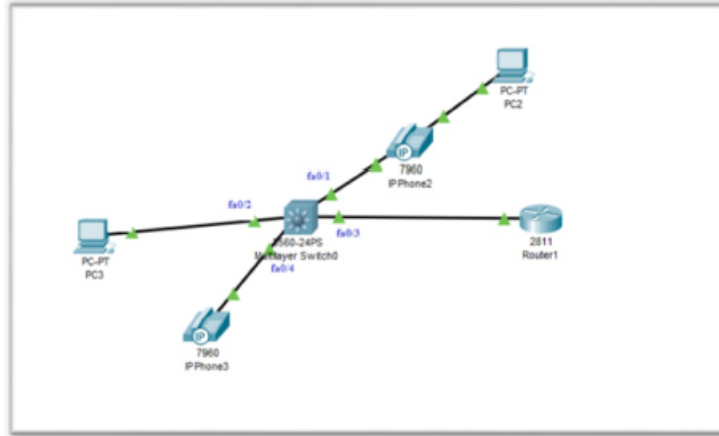


Figure 8.12. *Example of implementation of a VoIP solution*

QoS must be comprehensively implemented:

```
Switch(conf)#mlsqos
```

Next, we configure the fao/1 interface, where first the IP telephone is connected and then the PC:

```
Switch(config)#interface fao/1
```

```
Switch(config-if)#switchport mode access
```

```
Switch(config-if)#switchport access vlan 20
```

The priority (COS) announced by the telephone will be accepted:

```
Switch(config-if)#mlsqos trust cos
```

The following command is used to instruct the Cisco IP telephone to tag data traffic coming from a PC connected to the same port with a predetermined COS value. This is in order to prevent the PC from tagging its frames with the VLAN voice tag and thus not assigning the same priority to voice and data traffic:

```
Switch(config-if)#switchport priority extend cos 4
```

For the Fao/4 interface, which includes an IP telephone only, we use the following commands:

```
Switch(config)#interface fao/4
```

```
Switch(config-if)#switchport mode access
```

```
Switch(config-if)#switchport voice vlan 10
```

```
Switch(config-if)#mlsqos trust cos
```

To verify our configuration, we use the following command:

```
Switch#show interfaces fao/1 switchport
```

Name: Fa0/1

Switchport: Enabled

Administrative Mode: static access

Operational Mode: static access
Administrative Trunking Encapsulation: dot1q
Operational Trunking Encapsulation: native
Negotiation of Trunking: Off
Access Mode VLAN: 20 (data)
Trunking Native Mode VLAN: 1 (default)
Voice VLAN: 10
Administrative private-vlan host-association: none
Administrative private-vlan mapping: none
Administrative private-vlan trunk native VLAN: none
Administrative private-vlan trunk encapsulation: dot1q
Administrative private-vlan trunk normal VLANs: none
Administrative private-vlan trunk private VLANs: none Operational private-vlan: none
Trunking VLANs Enabled: All
Pruning VLANs Enabled: 2-1001
Capture Mode Disabled
Capture VLANs Allowed: ALL
Protected: false
Unknown unicast blocked: disabled
Unknown multicast blocked: disabled
Appliance trust: 4

Since this is a simulation with a packet tracer, verification of the prioritization of voice traffic by the IP telephone (Cisco phone) has not been successful. In fact, IP telephones do not transmit prioritized frames, as shown by the content of the frames exchanged between the IP telephone and the switch in packet tracer simulation mode in [Figure 8.13](#). The frames preserve the default COS and DSCP values that correspond to the best effort model. This may be due to bugs in this software.

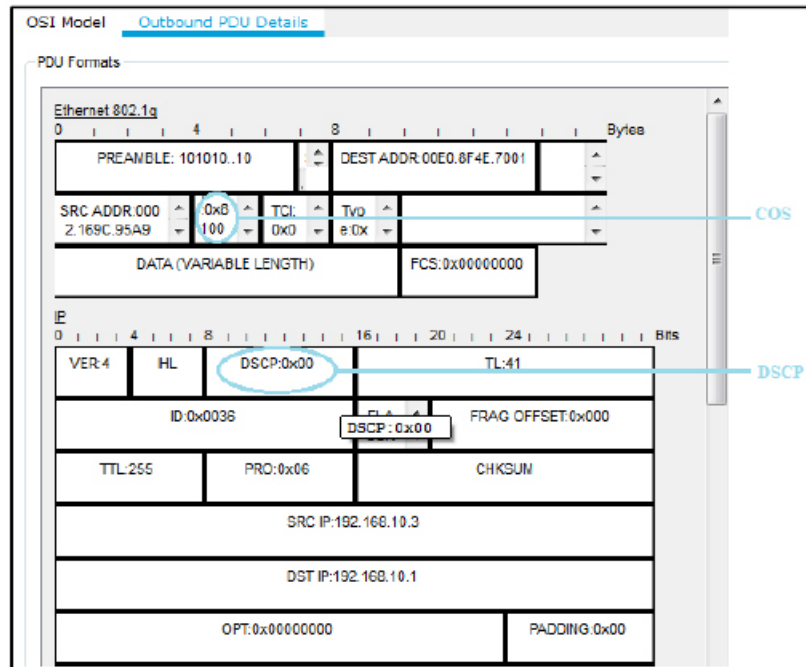


Figure 8.13. Failure of prioritized frame transmission. For a color version of the figure, see www.iste.co.uk/helali/systems.zip

To rectify this situation, a configuration based on the MQC approach, which consists of defining a traffic class, creating a corresponding traffic policy, and attaching it to an interface, can be applied.

Configuring flow classes

We will first define two access control lists:

```
Switch(config)# ip access-list extended voice
Switch(config-std-nacl)#permit ip 192.168.10.0 0.0.0.255 any
Switch(config-std-nacl)#ip access-list extended others
Switch(config-ext-nacl)#permit tcp any any eq 1521
Switch(config-ext-nacl)#permit tcp any any eq 1810
Switch(config-ext-nacl)#permit tcp any any eq 2481
Switch(config-ext-nacl)#permit tcp any any eq 7778
```

Next, two classes of service will be configured:

```
Switch(config)#class-map classvoice
Switch(config-cmap)#match access-group name voice
Switch(config-cmap)#exit
Switch(config)#class-map pcs
Switch(config-cmap)#match access-group name others
```

Creating the policy to be adopted:

```
Switch(config)#policy-map test
Switch(config-pmap)#class classvoice
Switch(config-pmap-c)#set precedence 7
Switch(config-pmap-c)#exit
Switch(config-pmap)#class pcs
Switch(config-pmap-c)#set dscp af21
```

Attaching the policy created to the interface

The policy must be applied at the interface entry, in our case fao/1 and fao/4:

```
Switch(config)#int fao/1
Switch(config-if)#service-policy input test
Switch(config)#int fao/4
Switch(config-if)#service-policy input test
```

Now, the IP telephones are transmitted prioritized frames as shown by the content of the frames exchanged between the IP telephone (IP Phone 3) and the switch in packet tracer simulation mode in [Figure 8.14](#).

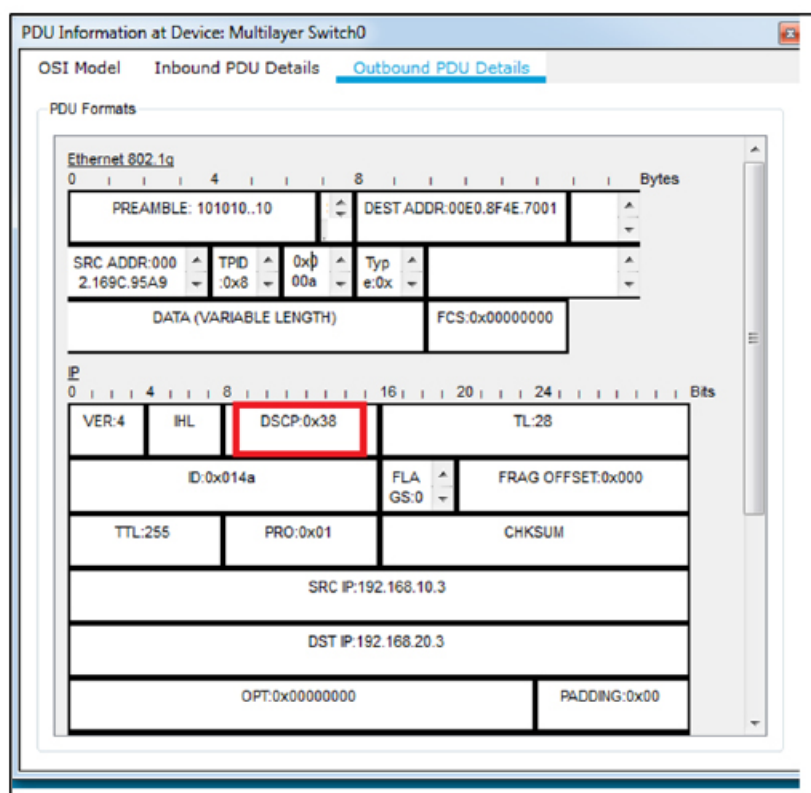


Figure 8.14. Successful transmission of prioritized frames

The DSCP value is equal to 0x38, which gives in decimal points the value 56. This corresponds to the class CS7 that we have defined.

8.3. High availability

High availability can be defined as a system's capacity to ensure the provision of a service during a given period.

It represents all of the methods and means that contribute to ensuring that an IT infrastructure always provides the planned functionalities and services. Redundancy, backup, data replication and load distribution help to ensure high availability of the IT system. This high availability helps to ensure greater reliability and increased security, allowing the company to provide a higher SLA and thus higher QoS.

Table 8.8. *Rates of availability in nines*

Rate of availability	Duration of non-availability (per year)
99%	3 days and 15 hours
99.9%	8 hours and 48 minutes
99.99%	53 minutes
99.999%	5 minutes
99.9999%	32 seconds

The high availability approach with its different facets should be managed and studied from the design phase onward. The implementation of highly available infrastructures requires synergy among the design, procedures and technologies deployed. In this context, certain key questions must be asked about each network infrastructure, including the parts of the network in which high availability is required, and the parts in which it is not.

As a general rule, the datacenter, server area and backbone network are particularly vital in terms of availability. New methods such as link aggregation, router redundancy protocols and load balancing enable improved availability thanks to automatic malfunction tolerance (redundancy) and the quick switching time that comes with load balancing.

Redundancy is a procedure based on the multiplication of various components of an IT infrastructure (electrical energy sources, climate-control systems, network equipment, storage solutions, etc.). The goal is to ensure that it will function in the event that one or more of these components break down, thus guaranteeing its high availability. It can be deployed in all layers of the OSI model, from the physical layer to the application layer.

8.3.1. *Redundancy in the physical layer*

This is a matter of multiplying hardware and cabling. The fact of having multiple interfaces available on a piece of equipment, such as teaming for network cards, multiple sources of electrical power, and RAID hard disks, are some examples of this.

8.3.2. Redundancy in the data link layer

8.3.2.1. STP protocol

Designed for switches and standardized with the norm IEEE802.1D, this protocol is used to determine a loop-free path in a switched and redundant physical topology, that is, with loops that cause broadcast storms. It detects and deactivates real topology loops, thus providing a logical topology in the form of a tree. If a link fails, a new path is recalculated and the network is functional again.

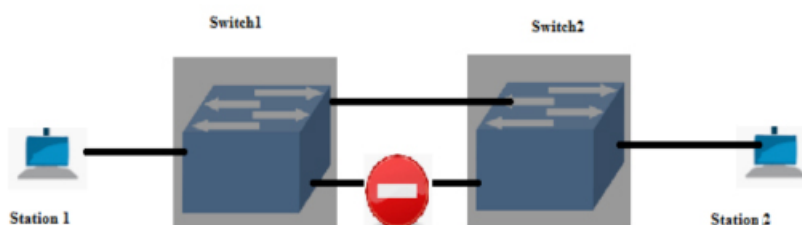


Figure 8.15. Principle of the STP protocol

8.3.2.2. EtherChannel

EtherChannel is a technology used to aggregate links or ports provided mainly by Cisco switches. It groups several physical links into a single logical link in order to improve bandwidth and to provide a level 2 redundancy mechanism. During normal operation, the load is balanced across all interfaces and, in the event of malfunction, data transport is confined to the remaining lines.

An EtherChannel link is composed of between two and eight active links which can be 100 Mbit/s, 1 Gbit/s or 10 Gbit/s, and possibly between one and eight inactive links in reserve for the event that the active links become unavailable. Indeed, theoretically, an aggregation of 4 100 Mb/s ports supplies a logical port with an output of 400 Mb/s.

The implementation of EtherChannel technology must fulfill the following conditions:

- equipment interfaces must support EtherChannel functionality;
- the same duplex mode (full duplex/half duplex) on both sides;
- the same speed on both sides;
- the same VLAN range authorized on both sides.

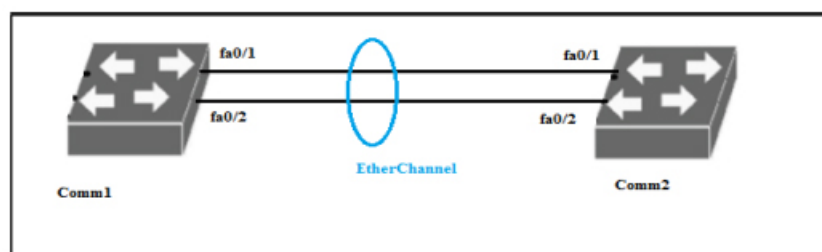


Figure 8.16. EtherChannel operation

EtherChannel technology is configured in two ways

The first method is purely manual; the appropriate commands must be entered on each port to be aggregated, on both sides of the link.

The second method is automatic and based on one of the two Cisco protocols: specifically, the Port Aggregation Protocol (PAgP) and the LACP protocol (Link Aggregation Control Protocol). At this stage, the existence of other protocols from other manufacturers should be indicated; in this case, Nortel MLT (Multi Link Trunking).

Example of manual configuration

The objective is to create an EtherChannel of two comm1 and comm2 switches by aggregating the two ports fa0/1 and fa0/2.

We will begin with the Comm1 switch:

```
Comm1(config)#interface fastethernet 0/1
Comm1(config-if)#port group 1 distribution destination
Comm1(config-if)#interface fastethernet 0/2
Comm1(config-if)#port group 1 distribution destination
Comm1(config-if)#exit
```

The Comm2 switch is configured as follows:

```
Comm2(config)#interface fastethernet 0/1
Comm2(config-if)#port group 1 distribution destination
Comm2(config-if)#interface fastethernet 0/2
Comm2(config-if)#port group 1 distribution destination
Comm2(config-if)#exit
```

Example of configuration with the PAgP and LACP protocols

The network topology is shown in [Figure 8.17](#).

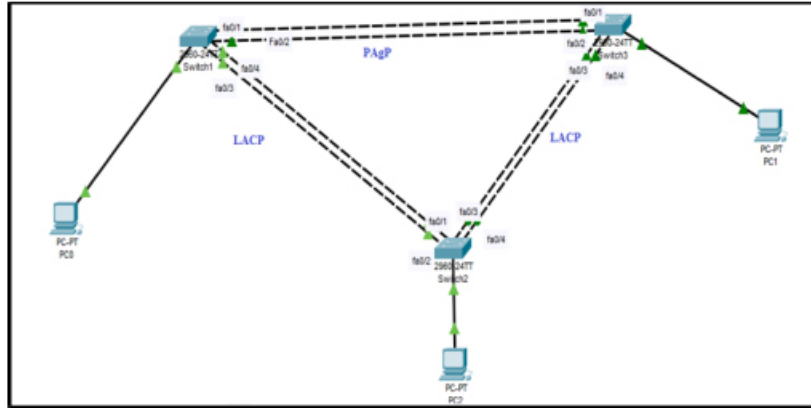


Figure 8.17. Mock LACP and PAgP protocol configurations

The PAgP protocol is a proprietary Cisco protocol referenced with IEEE802.3ad which offers three modes: on, desirable and auto:

- on mode activates the aggregation without using a negotiation protocol;
- desirable mode makes the aggregation request from the switch pair;
- auto mode waits for the negotiation and does not send requests, but uses the negotiation to form the aggregation.

Table 8.9. PAgP protocol modes

PAgP mode	ON	DESIRABLE	AUTO
ON	✓	X	X
DESIRABLE	X	✓	✓
AUTO	X	✓	X

The PAgP protocol, a level 2 redundancy protocol, is used to set up the EtherChannel technology between switch 1 and switch 3. The PAgP configuration between switch 1 and switch 3 is created using the following commands:

Switch1(config) #interface range fa0/1-2

Switch1(config-if-range)# channel-group 1 mode desirable

Switch1(config-if-range)#no shutdown

Switch3(config)#interface range fa0/1-2

Switch3(config-if-range)# channel-group 1 mode desirable

Switch3(config-if)#no shutdown

Switch1(config-if)#interface port-channel 1

Switch1(config-if)#switchport mode trunk

Switch1(config-if)#switchport trunk native vlan 99

```
Switch3(config)#int port-channel 1
```

```
Switch3(config-if)#switchport mode trunk
```

```
Switch3(config-if)#switchport mode native vlan 99
```

LACP is a standardized protocol developed by the IEEE. It functions like PAgP but supports various manufacturers. It has three configuration modes:

- ON mode serves, as in the case of PAgP, to activate the aggregation without using a negotiation protocol;
- active mode sends a request to the corresponding switch in order to create the aggregation;
- passive mode is characterized by a wait for the negotiation before switching toward aggregation. The compatibility between these different modes is shown in [Table 8.10](#).

Table 8.10. *LACP protocol modes*

LACP mode	ON	ACTIVE	PASSIVE
ON	✓	x	x
ACTIVE	x	✓	✓
PASSIVE	x	✓	x

The LACP protocol is configured between switch 1 and switch 2 and then between switch 2 and switch 3.

We begin by configuring LACP between switch 1 and switch 2:

```
Switch1(config)#interface range fa0/3-4
```

```
Switch1(config-if-range)#switchport mode trunk
```

```
Switch1(config-if-range)#switchport trunk native vlan 99
```

```
Switch1(config-if-range)#channel-group 3 mode active
```

```
Switch2(config)#interface range fa0/1-2
```

```
Switch2(config-if-range)#switchport mode trunk
```

```
Switch2(config-if-range)#switchport trunk native vlan 99
```

```
Switch2(config-if-range)#channel-group 3 mode passive
```

Next, we configure LACP between switch 2 and switch 3:

```
Switch2(config)#interface range fa0/3-4
```

```
Switch2(config-if-range)#switchport mode trunk
```

```
Switch2(config-if-range)#switchport trunk native vlan 99
```

```
Switch2(config-if-range)#channel-group 3 mode active
```

```
Switch3(config)#interface range fa0/3-4
```

```
Switch3(config-if-range)#switchport mode trunk
```

```
Switch3(config-if-range)#switchport trunk native vlan 99
```

```
Switch3(config-if-range)#channel-group 3 mode passive
```

Verification that EtherChannel is functioning correctly is shown below:

```
Switch1#show etherchannel summary
```

Flags: D - down P - in port-channel

I - stand-alone s - suspended

H - Hot-standby (LACP only)

R - Layer3 S - Layer2

U - in use f - failed to allocate aggregator

u - unsuitable for bundling

w - waiting to be aggregated d - default port

Number of channel-groups in use: 2

Number of aggregators: 2

Group Port-channel Protocol Ports

-----+-----+-----+-----

1 Po1(SU) PAgP Fa0/1(P) Fa0/2(P)

2 Po2(SU) LACP Fa0/3(P) Fa0/4(P)

```
Switch3#show etherchannel summary
```

Flags: D - down P - in port-channel

I - stand-alone s - suspended H - Hot-standby (LACP only)

R - Layer3 S - Layer2

U - in use f - failed to allocate aggregator

u - unsuitable for bundling w - waiting to be aggregated

d - default port

Number of channel-groups in use: 2

Number of aggregators: 2

Group Port-channel Protocol Ports

-----+-----+-----+-----

1 Po1(SU) PAgP Fa0/1(P) Fa0/2(P)

3 Po3(SD) LACP Fa0/3(I) Fa0/4(I)

Switch2#show etherchannel summary

Flags: D - down P - in port-channel

I - stand-alone s - suspended

H - Hot-standby (LACP only)

R - Layer3 S - Layer2

U - in use f - failed to allocate aggregator

u - unsuitable for bundling

w - waiting to be aggregated

d - default port

Number of channel-groups in use: 2

Number of aggregators: 2

Group Port-channel Protocol Ports

-----+-----+-----+-----

2 Po2(SU) LACP Fa0/1(P) Fa0/2(P)

3 Po3(SD) LACP Fa0/3(I) Fa0/4(I)

8.3.3. Redundancy in the network layer

To ensure high availability, individual failure points must be eliminated. Thus, a network gateway must be split at least once to prevent a gateway malfunction from making it impossible to communicate with the whole network.

8.3.3.1. HSRP (Hot Standby Router Protocol)

This is a proprietary Cisco protocol that offers higher availability of the network gateway. A group of routers composed of at least two routers is the foundation stone of this protocol and is identified by a number ranging from 0 to 255.

However, a single virtual IP address is assigned to all routers belonging to the group, but each one has its own priority. The election process chooses the router with the highest priority or the highest IP address as the active router. The remaining routers are now in passive or standby mode, in which they do not transmit traffic but are listening to the active router to make sure it is still operational.

This is done by means of the heartbeat mechanism, in which the active router sends hello messages every three seconds. If the other routers do not receive a hello message

for 10 seconds, they initiate the process to reelect a new active router with the highest priority. This reelection process is transparent for users, who always have to communicate with a virtual router with the same IP address and the same associated MAC address, specifically 00:00:0C:07:AC:XX, in which the octet XX designates the HSRP group number.

This protocol can be configured at a router or a level 3 switch.

8.3.3.2. VRRP (Virtual Router Redundancy Protocol)

This protocol is very similar to the HSRP protocol and is in fact the standardized version of it. It is thus non-proprietary and compatible with other manufacturers. It uses a virtualization process to improve the default gateway availability of a local network.

A group of VRRP routers is composed of at least two routers, a master router and a backup router that share an IP address as a default gateway for members of the network. The associated MAC address is 00:00:5E:00:01:XX, where XX designates the VRID and varies from 0x01 to 0xFF. This VRID characterizes a group of VRRP routers and is the unique identifier of this group. The preemption parameter (preemp) enables the original master to take back the service of its replacement; in the event that it becomes functional again, it is activated by default. Hello messages are sent every 1 second. If, after 3 seconds, the backup routers have not received anything, they proceed with the election of a new master router.

8.3.3.3. GLBP (Gateway Load Balancing Protocol)

There is also a proprietary Cisco level 3 redundancy protocol based on the same basic principles as the HSRP and VRRP protocols described above. It also offers load balancing, which is fairly simple to configure.

The HSRP and VRRP protocols enable equitable load distribution via a specific configuration based on the use of several HSRP/VRRP groups with different active/master routers. For GLBP, this load distribution is based on weighting allowing for differentiated load balancing. To do this, it uses a virtual IP address associated with several virtual MAC addresses (four maximum) but belonging to the same GLBP group.

These MAC addresses have the form 00:07:B4:0X:XX:YY (in which the Xs designate the group number ranging from 1 to 1023 and the Ys represent the virtual router identifier). Grouping in this protocol is composed of two elements: the Active Virtual Gateway (AVG) router and the Active Virtual Forwarder (AVF) routers.

The elected AVG router has the highest priority or the highest IP address of the group. It replies to ARP requests generated by users, so as to distribute the load between the AVF routers based on load distribution algorithms (round robin, weighted or host dependent). It assigns virtual MAC addresses to the AVF routers and chooses from among them the standby AVC router, which replaces the AVG in the event of breakdown.

The AVF router simply forwards packets. The AVG router also forwards traffic. Hello messages are sent every 3 seconds, and the change of AVG router will take place if no hello message has been received after 10 seconds.

Table 8.11. Comparison between HSRP, VRRP and GLBP

	HSRP	VRRP	GLBP
Ownership	Cisco proprietary	Standardized	Cisco proprietary
>Roles	Active/passive	Master/backup	AVG/AVF
Virtual MAC address	00:00:0C:07:AC:XX	00:00:5E:00:01:XX	00:07:B4:0X:XX:YY
Hello time	3 s	1 s	3 s
Holding time	10 s	3 s	10 s
Load distribution	Manual (creation of HSRP groups)	Manual (creation of VRRP groups)	Dynamic and simple (load distribution algorithms)

Application: configuration of HSRP protocol

The network topology is shown in [Figure 8.18](#).

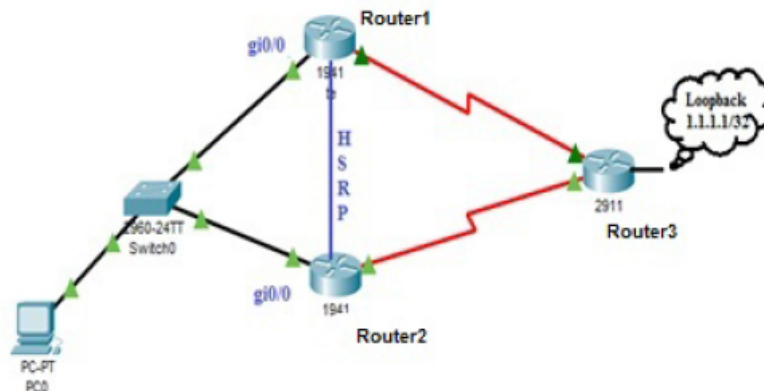


Figure 8.18. Mock configuration of the HSRP protocol. For a color version of the figure, see www.iste.co.uk/helali/systems.zip

Configuration of Router 1

The GigabitEthernet 0/0 interface is configured in HSRP group 1. It is assigned the virtual IP address 192.168.1.254. A priority of 250 (the highest priority is used to elect the active router) and the right of preemption is activated, that is, if Router 1 stops functioning, Router 2 takes over. If Router 1 becomes operational again, it resumes its role. In the event of nonpreemption, Router 2 will retain the role of active or gateway router. The real IP address of the g0/0 interface is 192.168.1.1:

```
Router1(config)#interface g0/0
Router1(config-if)#standby 1 ip 192.168.1.254
Router1(config-if)#standby priority 250
Router1(config-if)#standby 1 preempt
```

Configuration of Router 2

R2 is configured with a lower priority, so preemption activation is not necessary. The interface `gio/o` is allocated the real IP address `192.168.1.2`:

```
Router2(config)#interface gio/o
```

```
Router2(config-if)# standby 1 ip 192.168.1.254
```

We have configured a loopback interface at Router 3, `1.1.1.1 /32`, to simulate an Internet connection in order to test connectivity between PCO and the loopback interface.

To verify the implementation of the level 3 redundancy protocol, the `show standby` command is used:

```
Router1#show standby
```

```
GigabitEthernet0/0 - Group 1
```

```
State is Active
```

```
5 state changes, last state change 00:00:32
```

```
Virtual IP address is 192.168.1.254
```

```
Active virtual MAC address is 0000.0C07.AC01
```

```
Local virtual MAC address is 0000.0C07.AC01 (v1 default)
```

```
Hello time 3 sec, hold time 10 sec
```

```
Next hello sent in 0 secs
```

```
Preemption enabled
```

```
Active router is local
```

```
Standby router is 192.168.1.2, priority 100 (expires in 7 sec)
```

```
Priority 250 (configured 250)
```

```
Group name is hsrp-Gigo/o-1 (default)
```

```
Router2#show standby
```

```
GigabitEthernet0/0 - Group 1
```

```
State is Standby
```

```
9 state changes, last state change 00:00:50
```

```
Virtual IP address is 192.168.1.254
```

```
Active virtual MAC address is 0000.0C07.AC01
```

```
Local virtual MAC address is 0000.0C07.AC01 (v1 default)
```

```
Hello time 3 sec, hold time 10 sec
```

```
Next hello sent in 1.163 secs
```

```
Preemption disabled
```

```
Active router is 192.168.1.1, priority 250 (expires in 8 sec)
```

MAC address is 0000.0Co7.AC01

Standby router is local

Priority 100 (default 100)

Group name is hsrp-Gigo/o-1 (default)

The tracert command effectively tells us that the active router is indeed Router 1.

```
C:\>tracert 1.1.1.1

Tracing route to 1.1.1.1 over a maximum of 30 hops:

  1  1 ms    0 ms    0 ms    192.168.1.1
  2  1 ms    0 ms    2 ms    1.1.1.1

Trace complete.
```

We will now simulate a breakdown in Router 1 by deactivating its gio/o interface

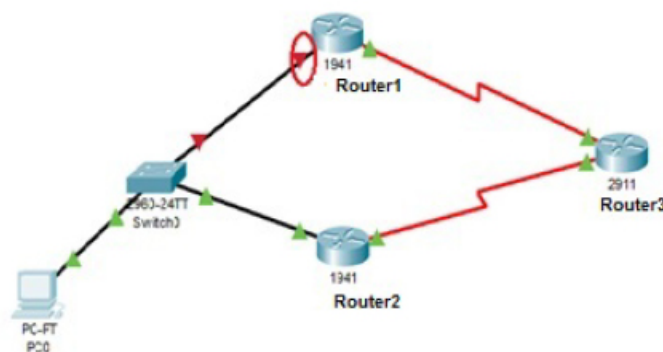


Figure 8.19. Simulation of a Router 1 breakdown. For a color version of the figure, see www.iste.co.uk/helali/systems.zip

The tracert command entered at PC0 effectively signals the switch toward the emergency router, in this case, Router 2.

```
C:\>tracert 1.1.1.1

Tracing route to 1.1.1.1 over a maximum of 30 hops:

  1  1 ms    0 ms    0 ms    192.168.1.2
  2  0 ms    0 ms    1 ms    1.1.1.1

Trace complete.
```

8.3.4. Redundancy in the application layer

Despite the use of redundancy in the hardware components of the network infrastructure, several factors may cause any service to become unavailable due to malfunctions in software, operating systems and applications. Thus, and in an effort to improve the overall availability of an IT system, applicative redundancy must be used for web servers, DNS servers or databases, for example.

To this end, methods have been developed such as load balancing, server switching and session switching. These are intended to relieve the burden on the server through a transparent distribution of communications over multiple computers, by subcontracting connection configuration actions and through verification of access

authorization in the upstream load balancer. For example, load balancing can be used with multiple web servers which all serve the same web pages and are grouped together into a cluster. A typical configuration is the formation of a pool or cluster of servers which is addressed by the terminals using a common virtual IP address. The load balancer converts this into the real address of the real server used. Load balancing helps to improve QoS, in that it enables the regulation of traffic, thus minimizing load peaks and avoiding problems of congestion.

Load balancing is now managed by hardware solutions as well as software solutions. Microsoft's NLB, HAProxy, Safekit and Linux's virtual server are some examples of software tools.

Another possibility for the implementation of redundancy is provided by virtualization. This makes multiple virtual servers available on a single physical machine, with the result that more cost-effective and eco-friendly redundancy has become possible.

Redundancy of different services must be used according to how vital they are to the IT system.

The benefits for network operators having to do with link aggregation, router redundancy protocols and load balancing in a network structure combined with layer 2 or 3 switching can justify added costs related to additional equipment and its setup. These benefits can be broken down as follows:

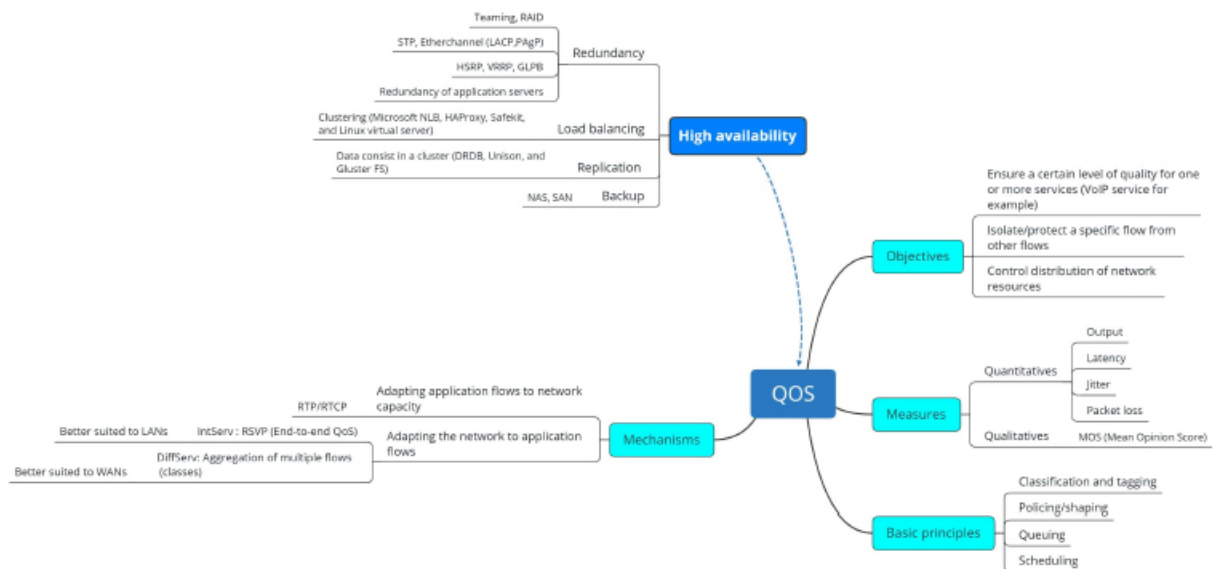
- tolerance of breakdowns with simultaneous load distribution;
- transparent redundancy for terminals and servers (VRRP and HSRP);
- good scalability of capacity within a sub-network;
- rapid switching in case of error;
- creation of appropriately sized broadcast domains;
- flexible configuration of sub-networks in buildings;
- compatibility with the use of standardized protocols, particularly between servers and switches;
- possibility of future implementation of QoS functions via classification and prioritization.

Replication plays a vital role in a cluster. It ensures data consistency in redundant servers and is intended to make data secure by synchronizing it among multiple servers. DRDB, Unison and Gluster FS are examples of replication software tools.

Redundancy, load balancing, and data replication are essential means to ensure high availability in an IT infrastructure. However, they cannot provide total protection against the risk of data loss. The data which presents the real capital of each company requires the establishment of a well studied storage and backup system on remote sites.

High availability requires the incorporation of multiple technologies which must work together to make sure that the IT system's services and data are functional on a virtually continuous basis. Several high availability solutions exist in today's market; we must simply find the one(s) best suited to our situation and requirements.

8.4. Chapter summary



Notes

This chart is also available at www.iste.co.uk/helali/systems.zip

¹ Quality measurement in IT networks, Razvan Beuran, 2004.